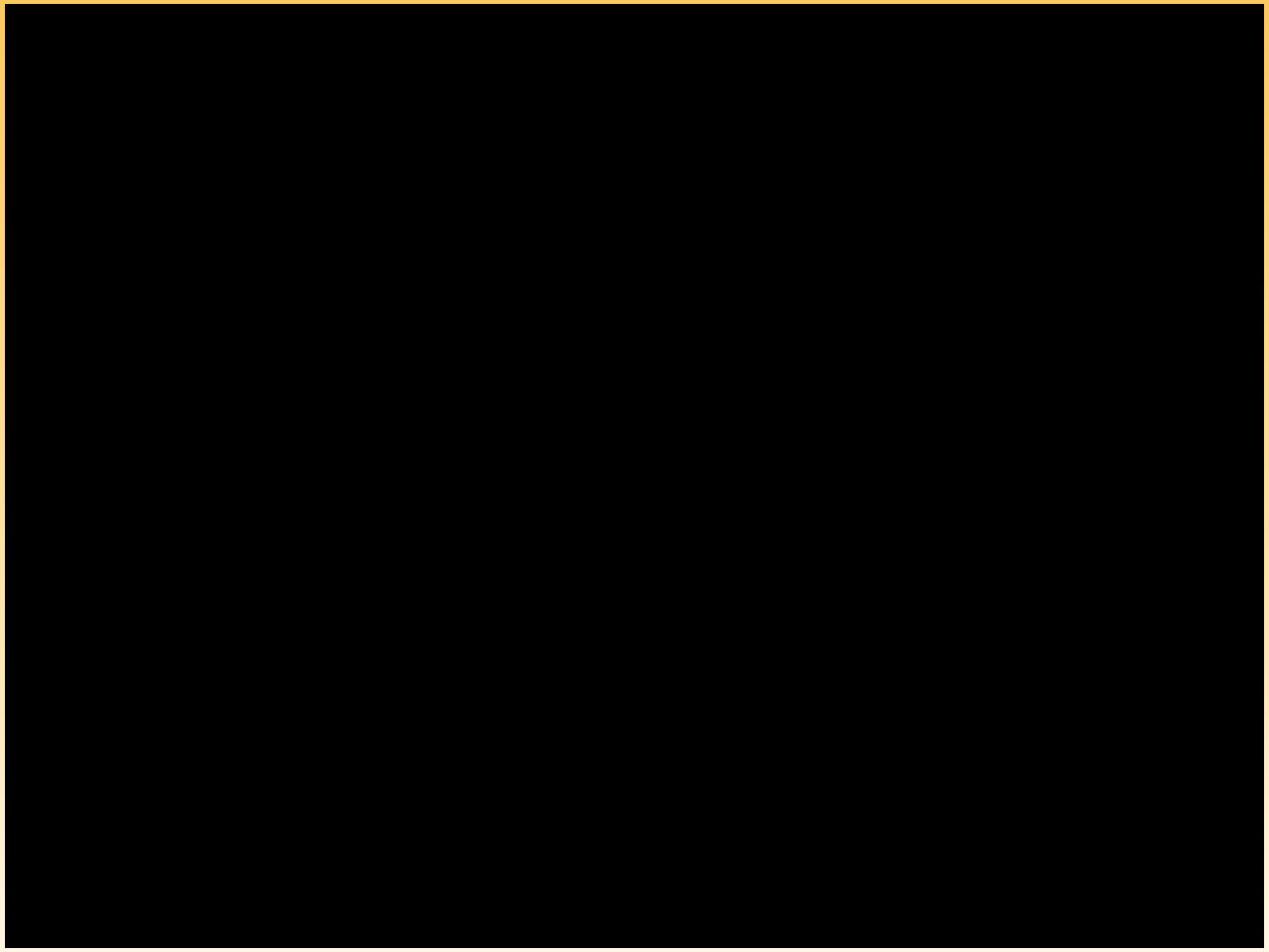
TCAS II, ACAS X, and ACAS Xu

CEC 300

EMBRY-RIDDLE
Aeronautical University

TCAS II

TCAS Video



TCAS II

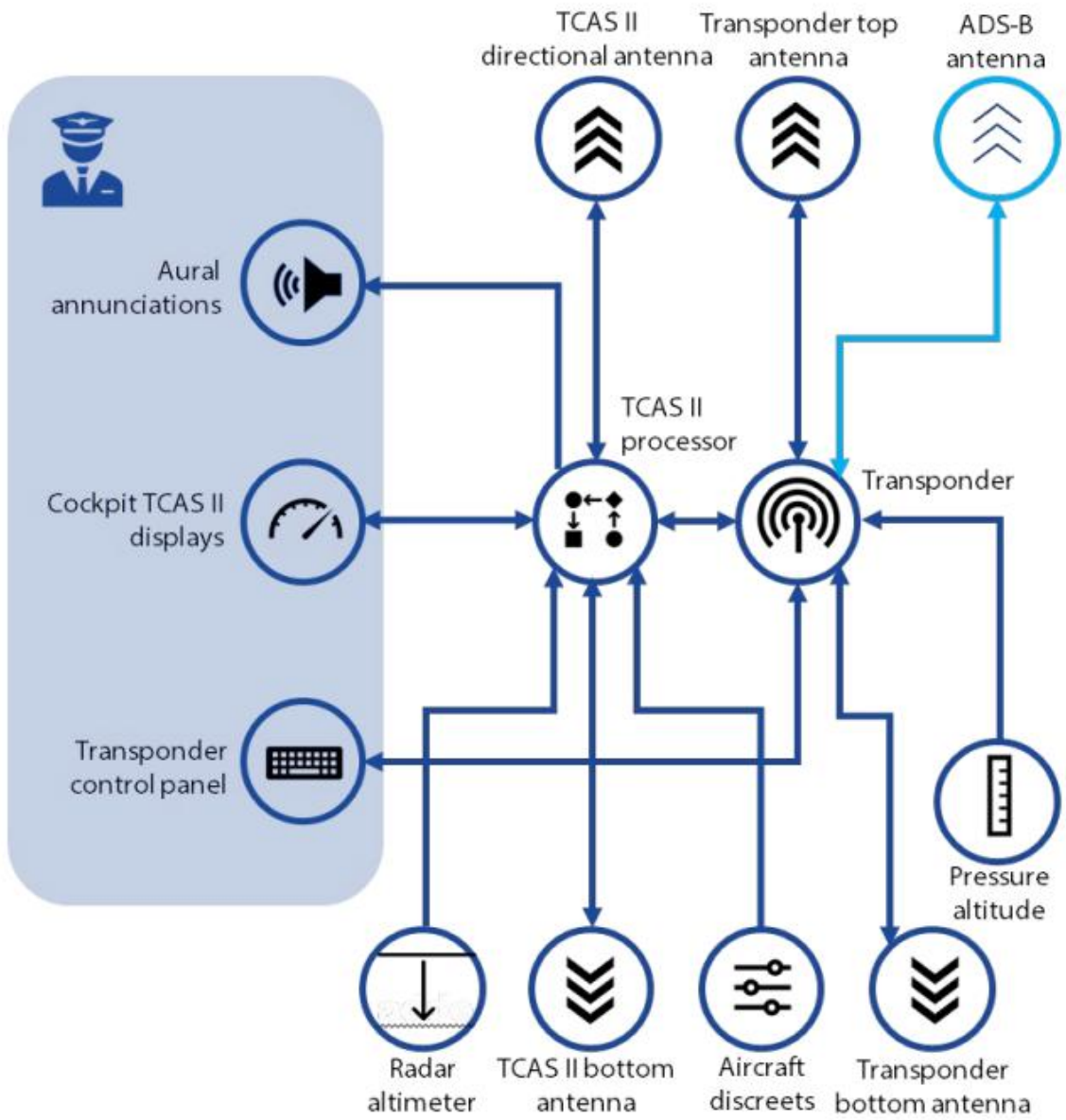


Figure 8: TCAS II installation schematic diagram.

EMBRY-RIDDLE
Aeronautical University

TCAS II

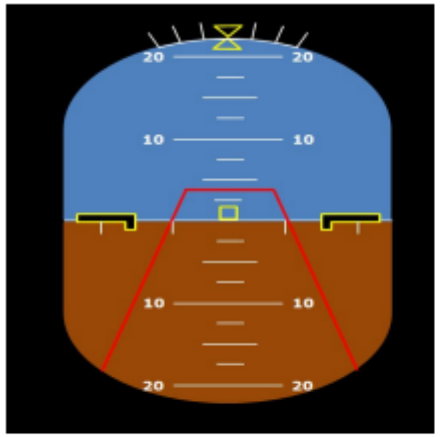


Figure 20: RA pitch cue on PFD.

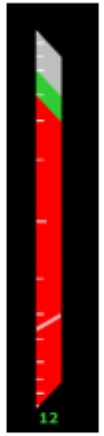


Figure 21: Vertical Speed Tape.

Primary Flight Display:

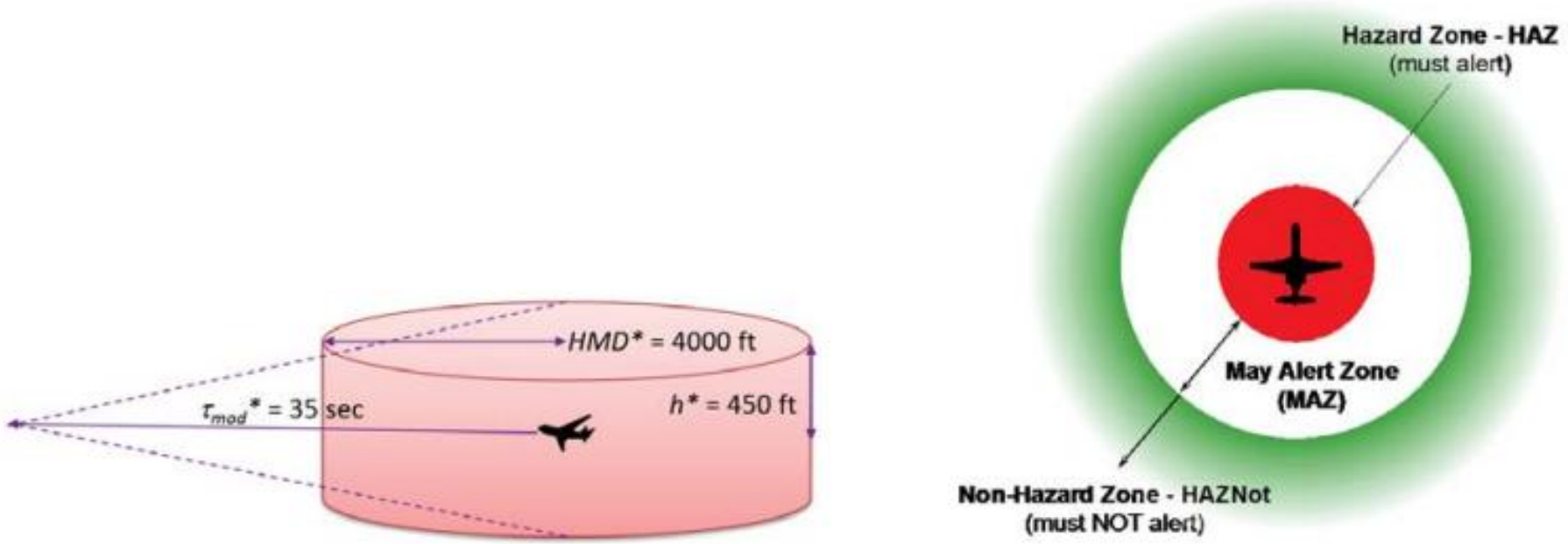
- RA pitch cue (left)
- Vertical speed tape (right)

Figure 13: Traffic display example – dedicated display.

Figure 14: Traffic display example – IVSI combined with traffic display.

Figure 15: Traffic display example – Electronic Flight Instrument System (EFIS).

TCAS II - Hazard Zone vs. Non-Hazard Zone



(a) DAA Well Clear Volume with En Route Parameter Values

(b) DAA Well Clear Hazard Regions

Fig. 1 Well Clear Volume and Alerting Zones

TCAS II (RA/TA thresholds)

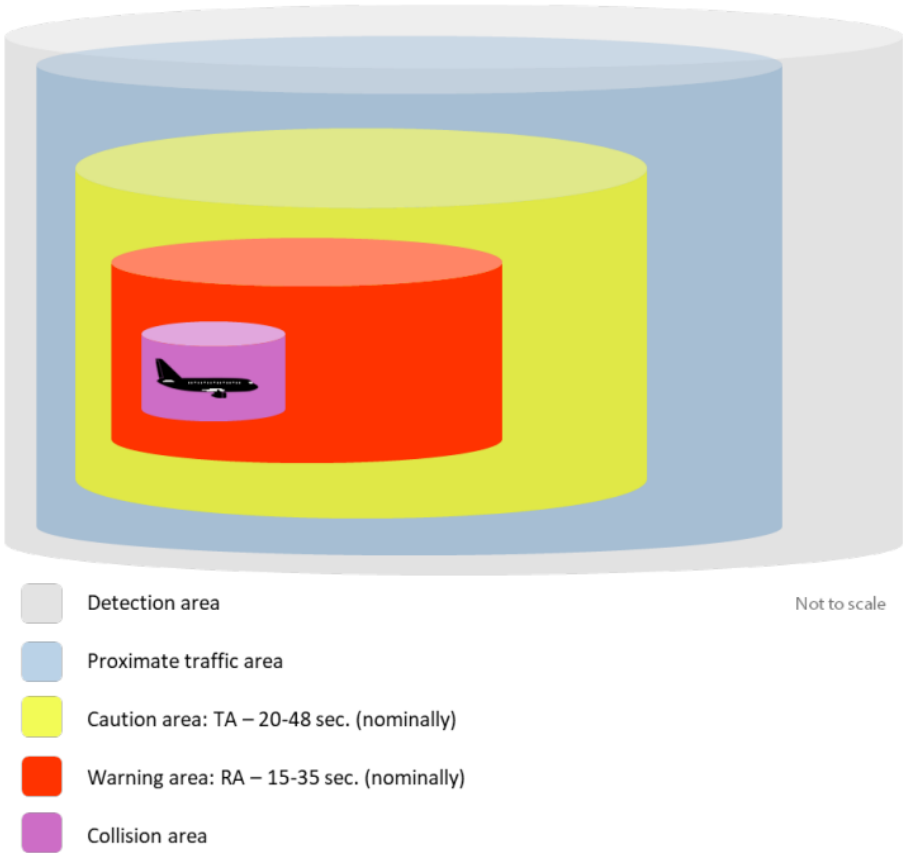


Figure 40: TCAS II protected volume (side and plan views).

Table 8: Alert thresholds related to altitude.

Own Altitude	*SL*	*tau* values (sec)		*TVTHR* (sec)	*DMOD* values (NM)		*ZTHR* (feet) Alt. Threshold		*ALIM* (feet)
		TA	RA	RA	TA	RA	TA	RA	RA
0 – 1000 ft AGL	2	20	no RA	no RA	0.30	no RA	850	no RA	no RA
1000 – 2350 ft AGL	3	25	15	15	0.33	0.20	850	600	300
2350 ft AGL – FL50	4	30	20	18	0.48	0.35	850	600	300
FL50 – FL100	5	40	25	20	0.75	0.55	850	600	350
FL100 – FL200	6	45	30	22	1.00	0.80	850	600	400
FL200 – FL420	7	48	35	25	1.30	1.10	850	700	600
Above FL420	7	48	35	25	1.30	1.10	1200	800	700

Time to Co-altitude Threshold. (**TVTHR**)

$$\tau \equiv -\frac{r}{\dot{r}},$$

$$\tau_{\mathrm{mod}} \equiv -\frac{r^2 - \mathrm{DMOD}^2}{r\dot{r}}$$

EMBRY-RIDDLE Aeronautical University

TA / RA times vs. altitude

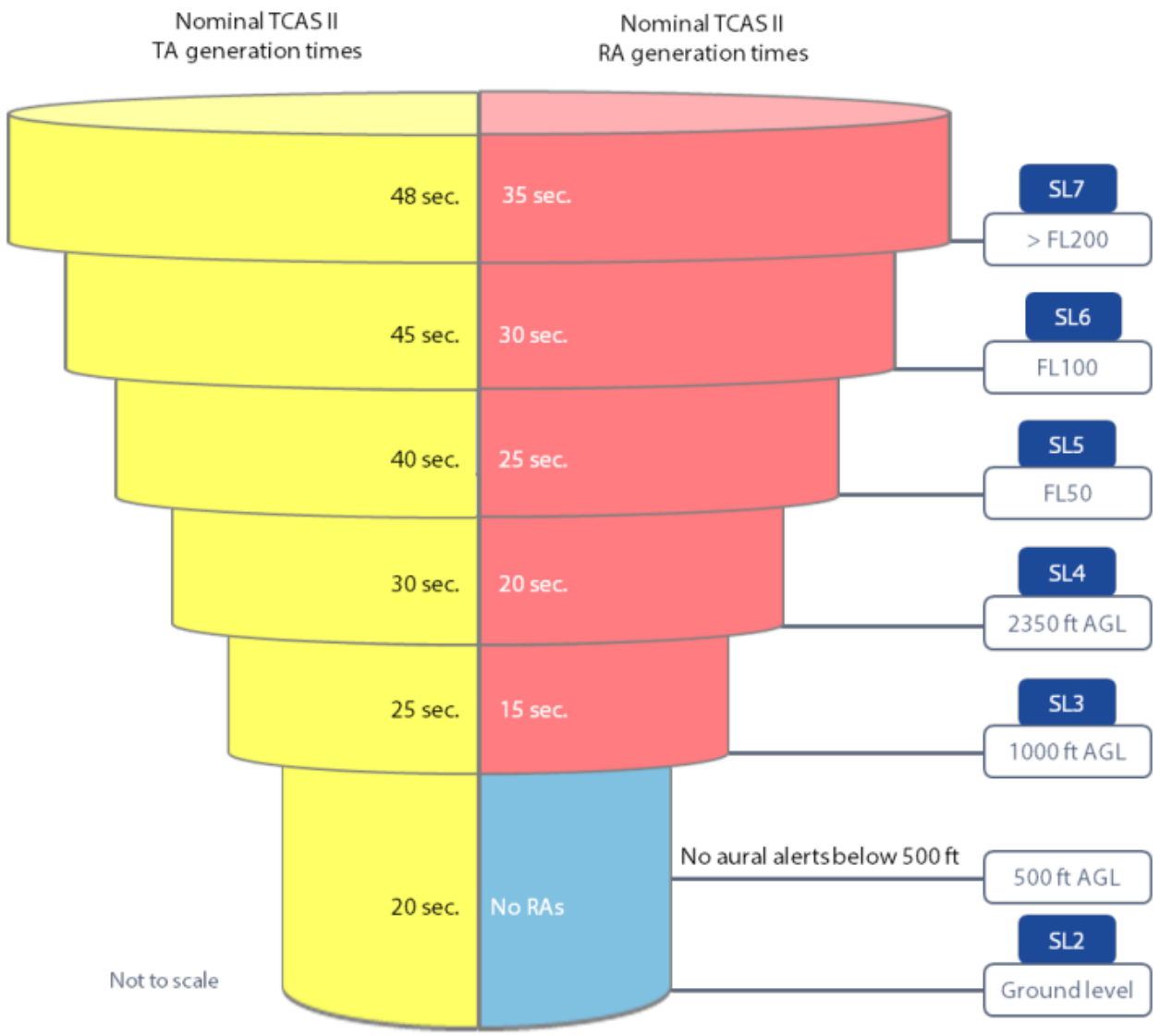
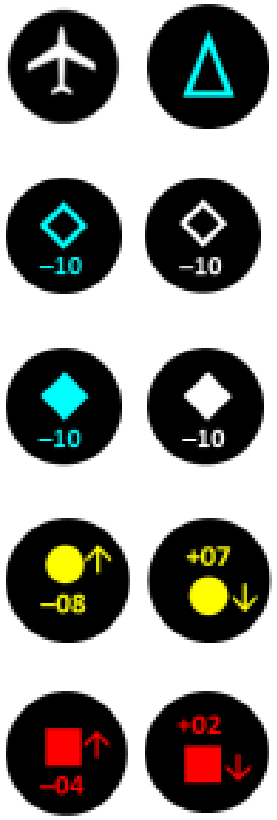


Figure 39: TCAS II nominal warning times and sensitivity levels.

EMBRY-RIDDLE Aeronautical University

TCAS II

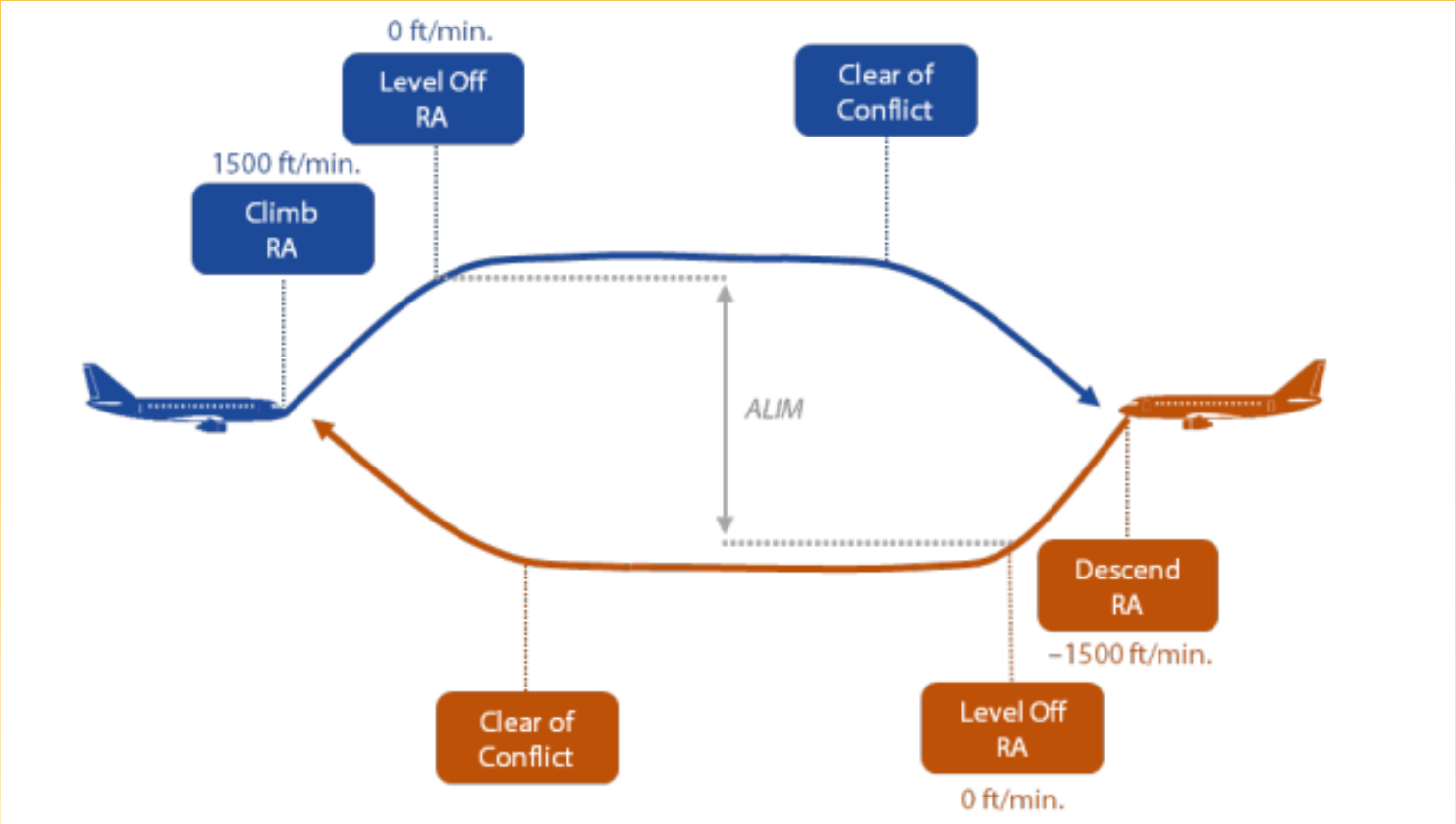


White or cyan (light blue) aircraft-like symbol or a triangle	Own aircraft
Hollow cyan (light blue) or white diamond	Other traffic
Solid cyan (light blue) or white diamond	Proximate traffic Aircraft within 6 NM and 1200 feet of own aircraft
Solid yellow or amber circle[48]	Traffic advisory Typically generated 20-48 seconds before CPA
Solid red square	Resolution advisory Typically generated 15-35 seconds before CPA

Figure 18: Traffic display symbology.

EMBRY-RIDDLE Aeronautical University

Advisories / Advisory Reversals



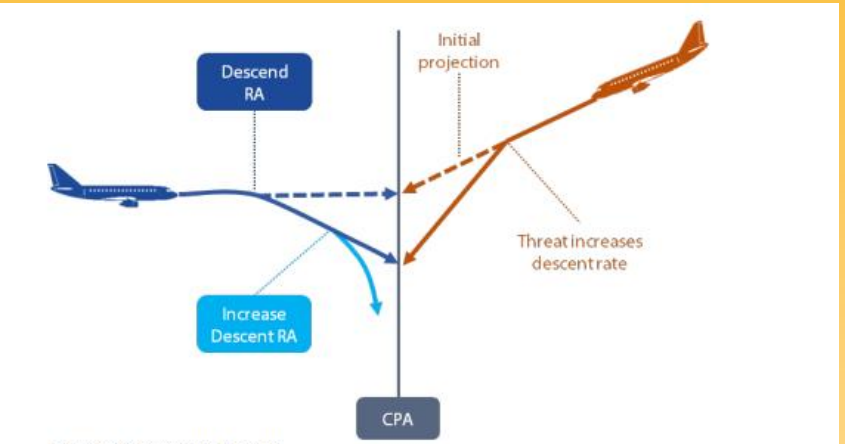


Figure 47: Increase vertical rate RA.

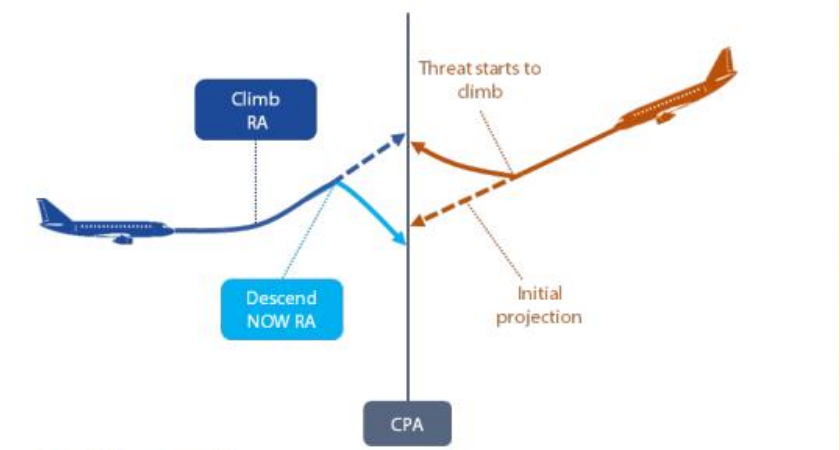
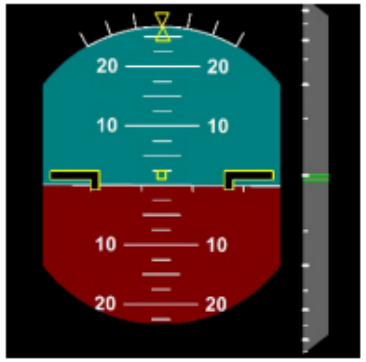
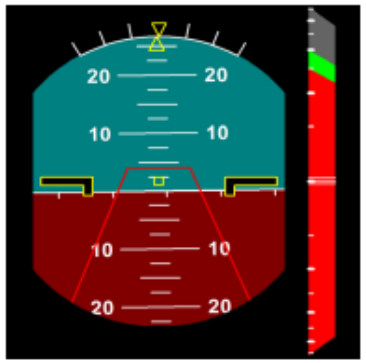
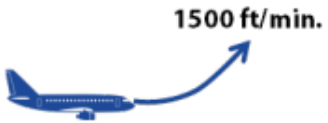
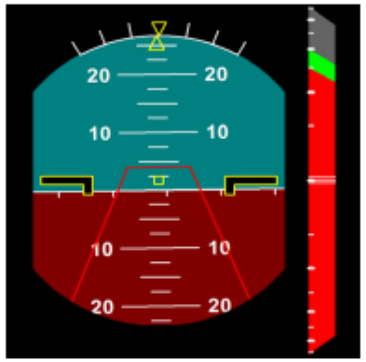
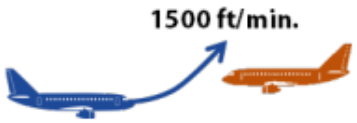


Figure 48: Sense reversal RA.

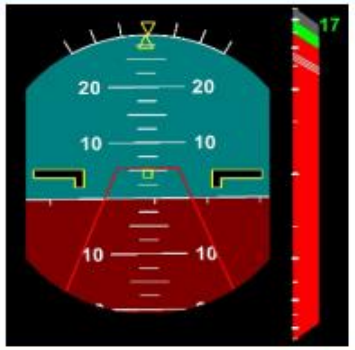
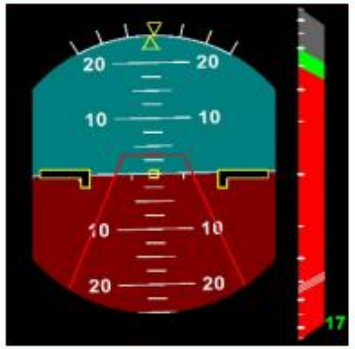
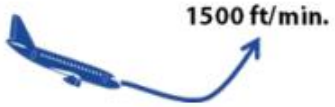
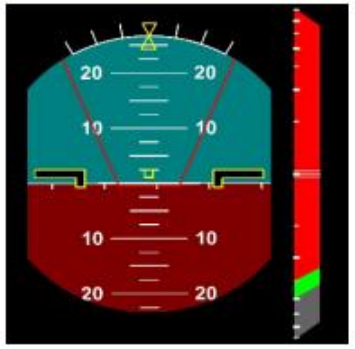
EMBRY-RIDDLE Aeronautical University

Single Threat Encounters

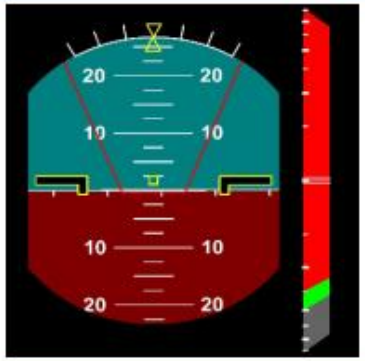
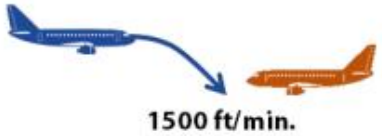
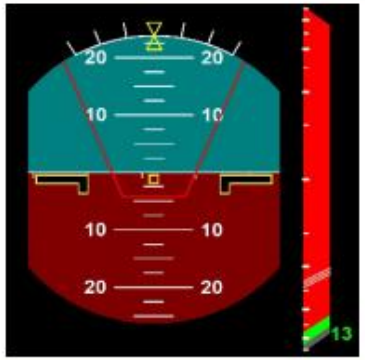
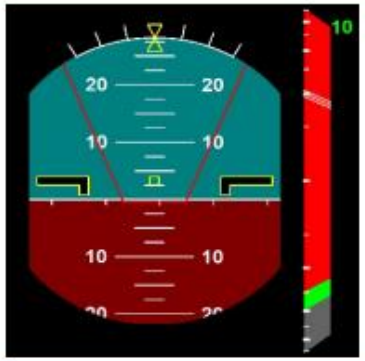
Table 2: TCAS II version 7.1 and ACAS Xa advisories aural annunciations and display as shown on (generic) IVSI and EFIS displays (single threat encounters).

Advisory	Aural annunciation	IVSI	EFIS	Manoeuvre
Traffic Advisory	Traffic, traffic TCAS II ver. 7.1			No manoeuvring
	Traffic, traffic ACAS Xa			
Climb	Climb, climb TCAS II ver. 7.1			
	Climb, climb ACAS Xa			
Crossing Climb	Climb, crossing climb; climb crossing climb TCAS II ver. 7.1			
	Climb, crossing climb; climb crossing climb ACAS Xa			

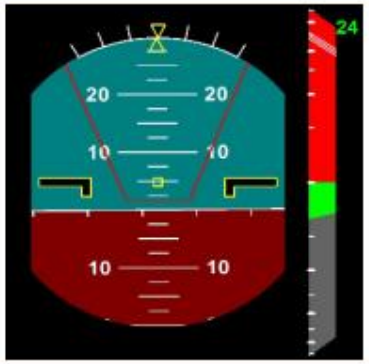
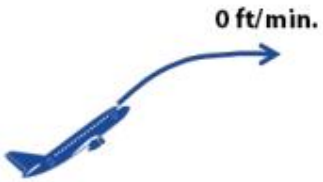
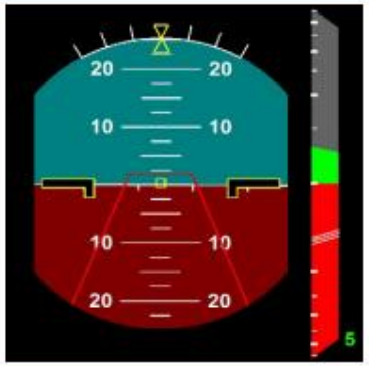
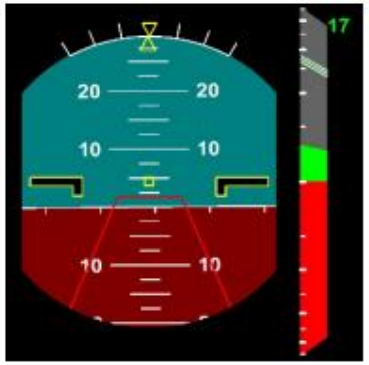
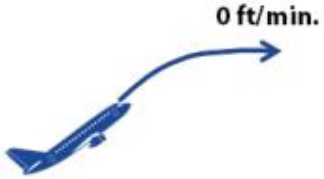
Single Threat Encounters

Advisory	Aural annunciation	IVSI	EFIS	Manoeuvre
Increase Climb Not possible as an initial RA	Increase climb, increase climb TCAS II ver. 7.1			
	Increase climb, increase climb ACAS Xa			
Reversal Climb Not possible as an initial RA	Climb, climb NOW TCAS II ver. 7.1			
	Climb, climb NOW ACAS Xa			
Descend	Descend, descend TCAS II ver. 7.1			
	Descend, descend ACAS Xa			

Single Threat Encounters

Advisory	Aural annunciation	IVSI	EFIS	Manoeuvre
Crossing Descend	Descend, crossing descend; descend crossing descend TCAS II ver. 7.1 Descend, crossing descend; descend crossing descend ACAS Xa			 1500 ft/min.
Increase Descent Not possible as an initial RA	Increase descend, increase descend TCAS II ver. 7.1 Increase descend, increase descend ACAS Xa			 2500 ft/min.
Reversal Descent Not possible as an initial RA	Descend, descend NOW TCAS II ver. 7.1 Descend, descend NOW ACAS Xa			 1500 ft/min.

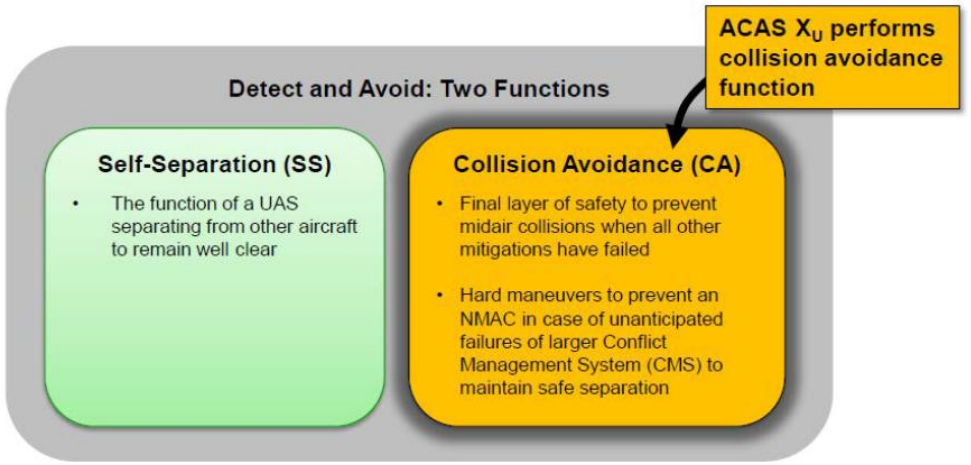
Single Threat Encounters

Advisory	Aural annunciation	IVSI	EFIS	Manoeuvre
Level Off (downward) Initial RA issued while climbing	Level off, level off TCAS II ver. 7.1 Level off, level off ACAS Xa			
Level Off (upward) Initial RA issued while descending	Level off, level off TCAS II ver. 7.1 Level off, level off ACAS Xa			
Level Off (downward) Weakening RA issued after climb Not possible as an initial RA	Level off, level off TCAS II ver. 7.1 Level off, level off ACAS Xa			

ACAS X_a

ACAS X

- Airborne Collision Avoidance System X
  - Updated standard for collision avoidance system, specialized by aircraft type.
- Differences with TCAS II
  - Use of ADS-B to create hybrid surveillance
  - Improved alerting and advisory logic
- ACAS Xu – Designed for UAS applications, RTCA DO-385



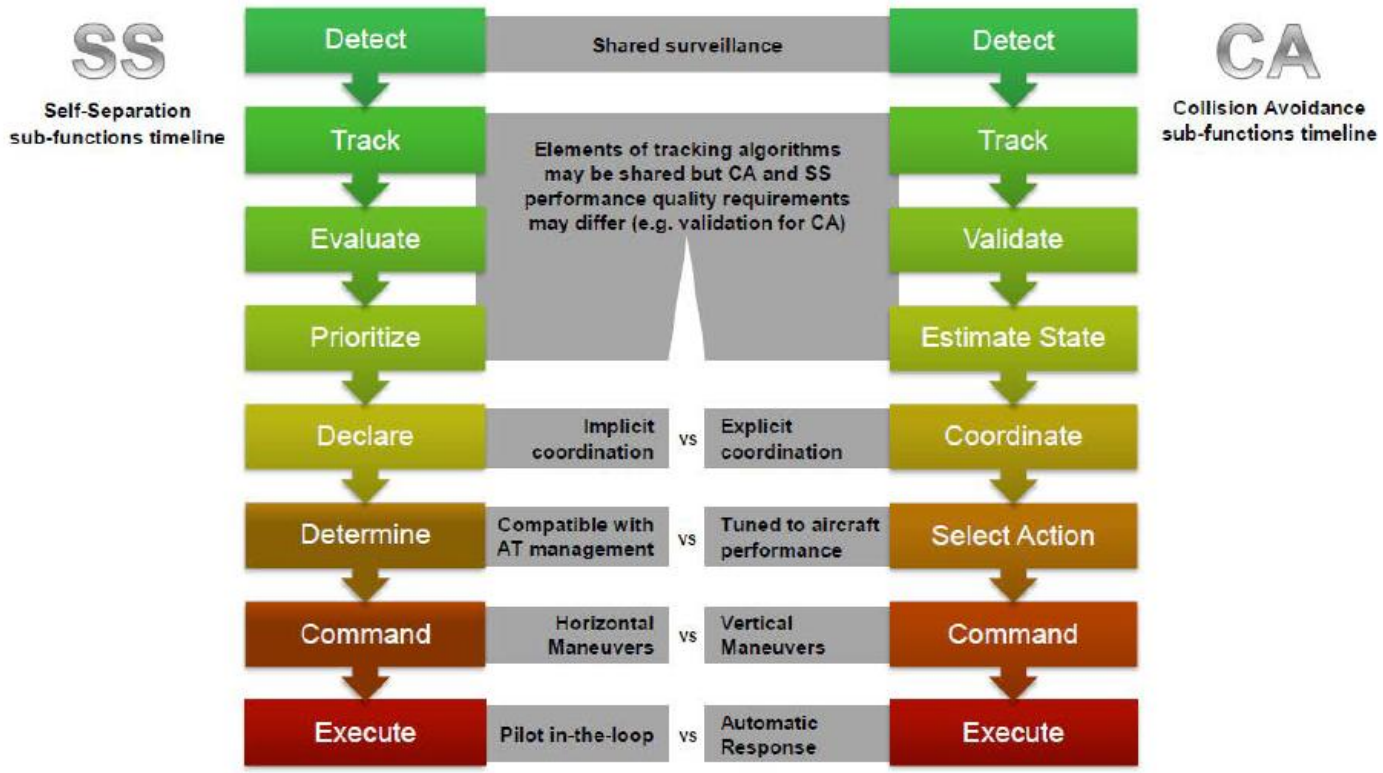


Figure 8: Comparison of Self-Separation vs. Collision Avoidance in the ACAS X context (Source: FAA [Ref 5])

RTCA SC-228 is looking to extend the ACAS Xu by defining remain well clear functional requirements.

Self-Separation vs. Collision Avoidance

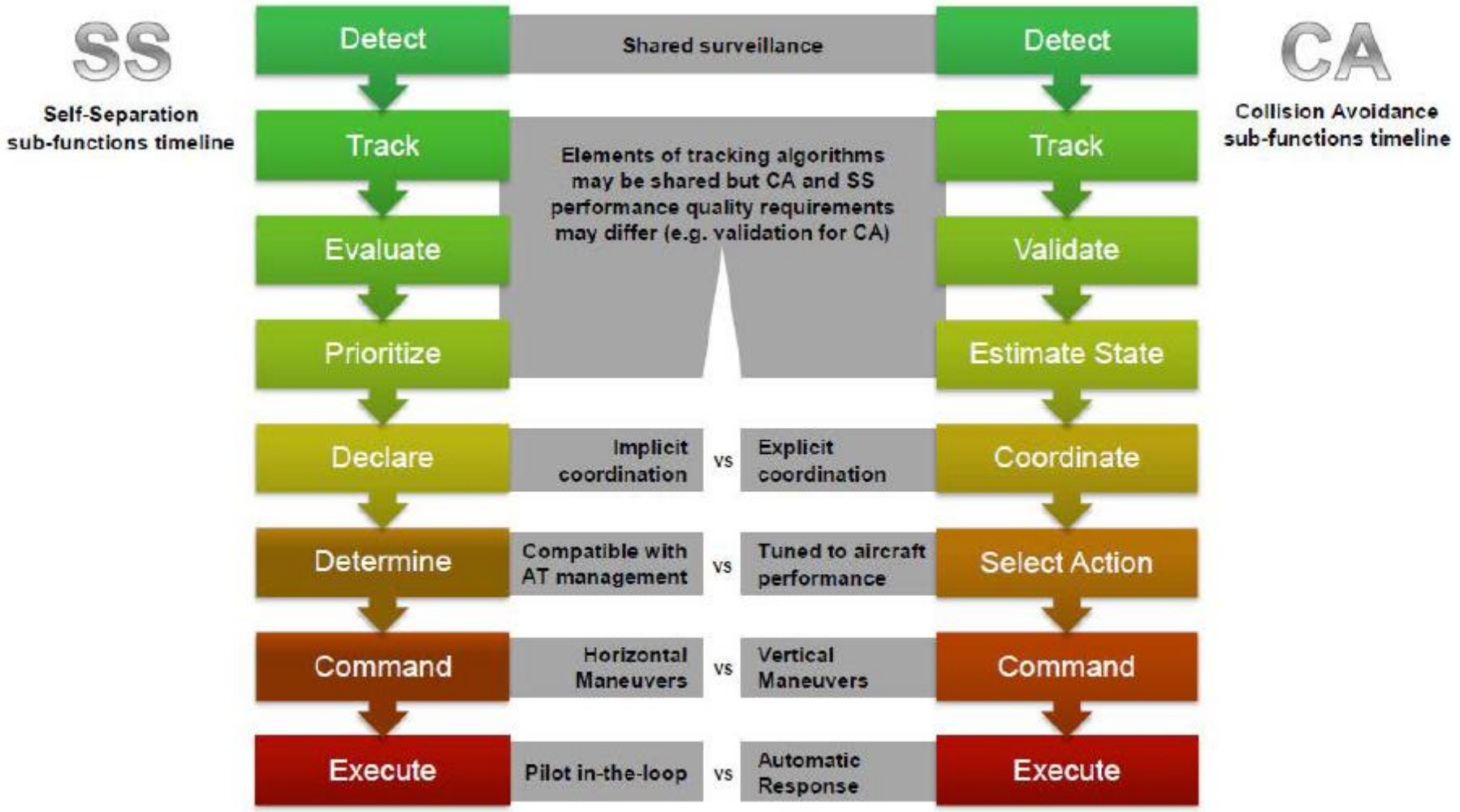


Figure 8: Comparison of Self-Separation vs. Collision Avoidance in the ACAS X context (Source: FAA [Ref 5])

ACAS X

Table 2: ACAS X variants and relevant MOPS.

	ACAS X system	Purpose	RTCA MOPS / EUROCAE MOPS
	Xa	The general purpose ACAS X Successor to TCAS II	DO-385A / ED-256A
	Xo	Specific operations, e.g., closely spaced parallel approaches	DO-385A / ED-256A
	Xu	For unmanned aircraft	DO-386 / ED-275
	sXu	For small unmanned aircraft (wingspan less than 15 m)	DO-396 / No equivalent
	Xr	For rotorcraft (helicopters)	Under development (estimated completion 2025)
	Xp	For general aviation (passive)	Potential future system

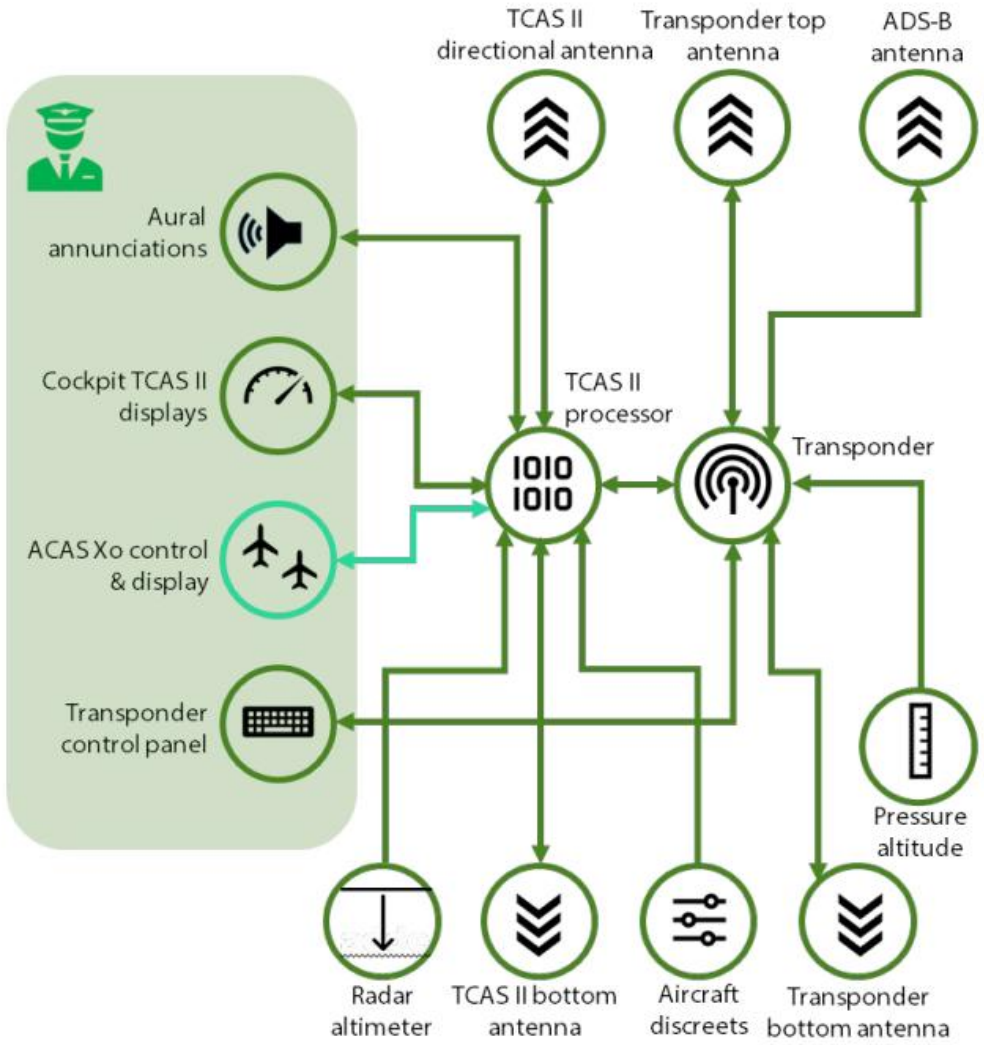


Figure 9: ACAS Xa installation schematic diagram.

ACAS Xa Logic

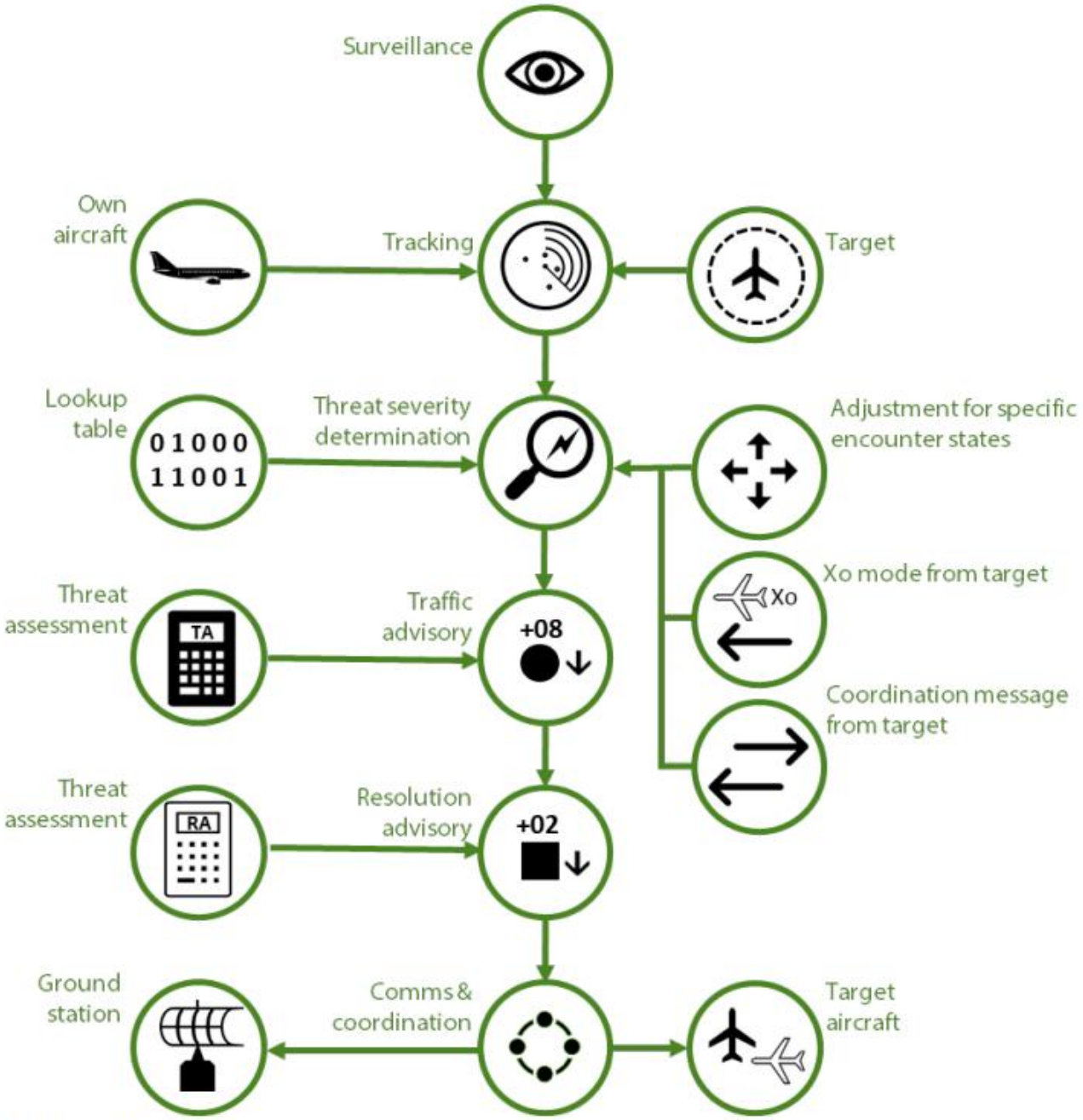


Figure 52: ACAS Xa logic functions.

Uncertainty and State Estimation

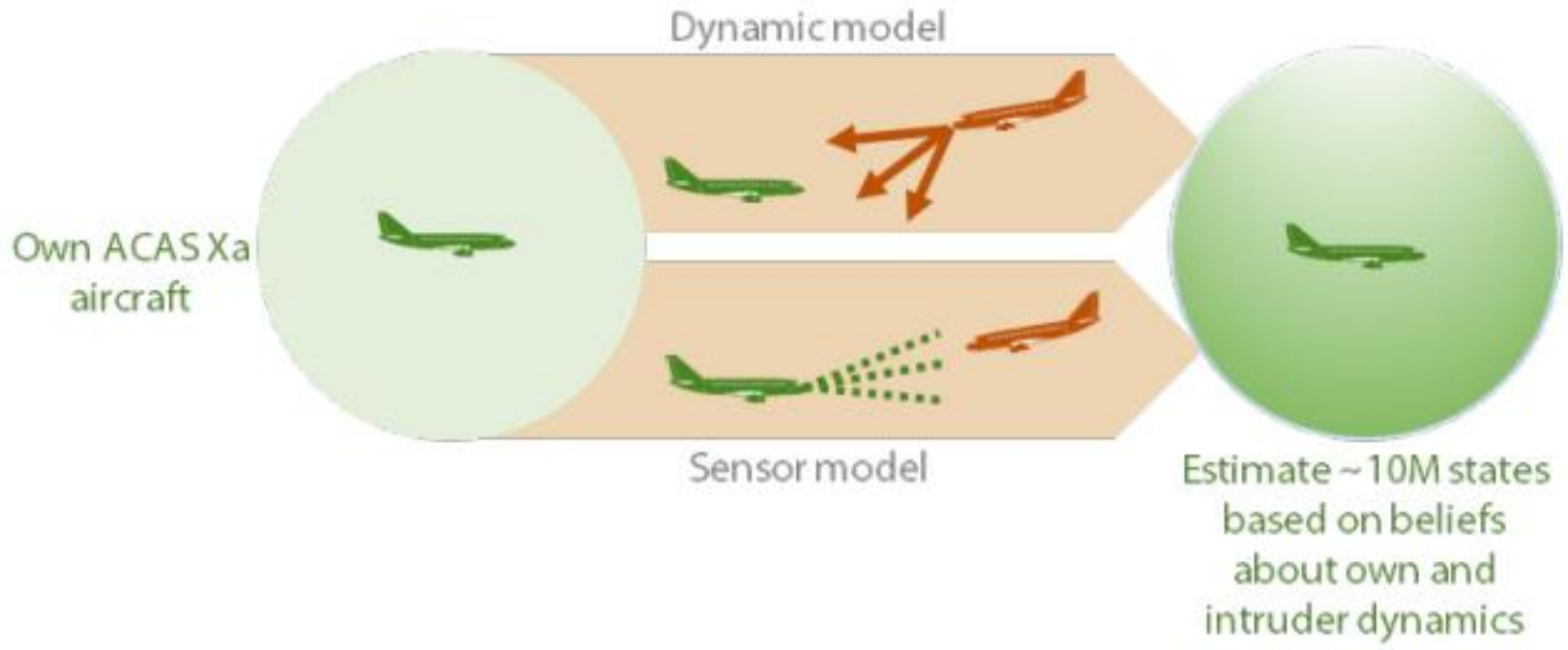


Figure 50: ACAS Xa state estimation.

ACAS Xa combines a probalistic dynamics model to determine where the aircrafts may go and a probabilistic sensor model to account for noise/error in sensor measurements.

Based on these models and the state of the vehicle, the aircraft's state can be approximated.

TCAS II vs. ACAS Xa

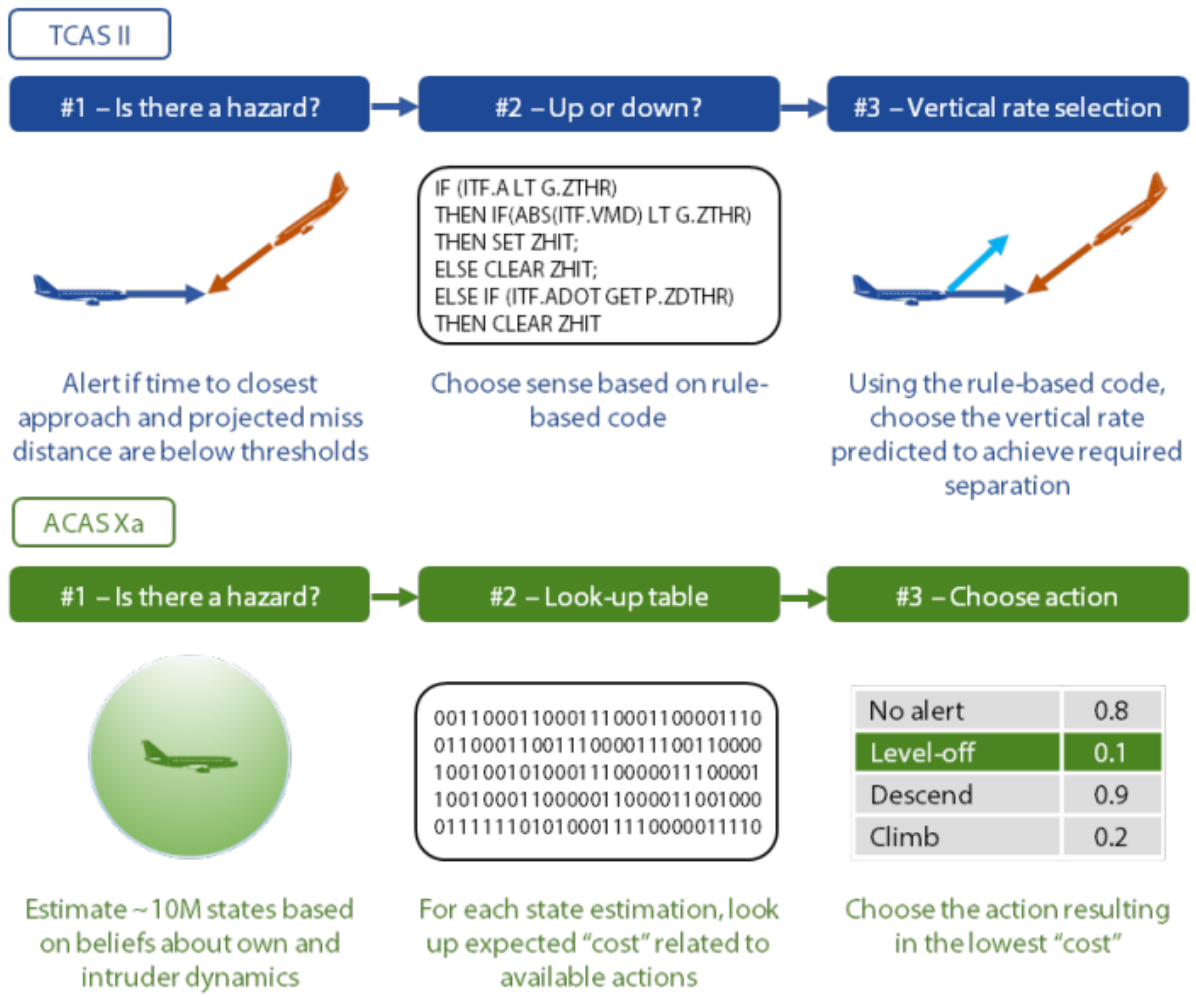


Figure 51: RA selection process: TCAS II vs. ACAS Xa.

- With TCAS II, a **set of rules** is used to determine given the states of ownship and intruder the type of traffic alert or resolution advisory to issue.
- ACAS Xa uses a **look-up table** to determine possible actions to take given the state.
  - Considering the **associated cost of the action**, the system selects **action with the lowest cost**.

Explainer: Markov Decision Process

ACAS X: Whole Airspace Protection

TCAS II vs. ACAS Xa – Advisory Types and Annunciations

Advisory	Aural annunciation	
	TCAS II ver. 7.1	ACAS Xa
Maintain Vertical Speed Issued while climbing	Maintain vertical speed, maintain	Climb, climb
Crossing Maintain Vertical Speed Issued while climbing	Maintain vertical speed, crossing maintain	Climb, crossing climb; climb crossing climb
Maintain Vertical Speed Issued while descending	Maintain vertical speed, maintain	Descend, descend
Crossing Maintain Vertical Speed Issued while descending	Maintain vertical speed, crossing maintain	Descend, crossing descend; descend, crossing descend
Limit Climb to 500, 1000 or 2000 ft/min.	Monitor vertical speed	—
Limit Descend to 500, 1000 or 2000 ft/min.	Monitor vertical speed	—
Do Not Climb Weakening RA Issued after maintain descend RA	—	Monitor vertical speed
Do Not Descend Weakening RA Issued after maintain climb RA	—	Monitor vertical speed
Multi-threat encounter Maintain existing vertical speed RA	Maintain vertical speed, maintain	Level off, level off
Multi-threat encounter Preventive Do Not Climb & Descend RA	Monitor vertical speed	—

EMBRY-RIDDLE Aeronautical University

ACAS Xo – handling special cases

- ACAS Xo – is a designation for special cases of operations.

- Pilot can designate another flight as:
  - **CSPO** – Closely spaced parallel approach
    - Resolution advisory thresholds reduced.
  - **DNA** - Designated No Alert
    - Aircraft is ignored, no alerts

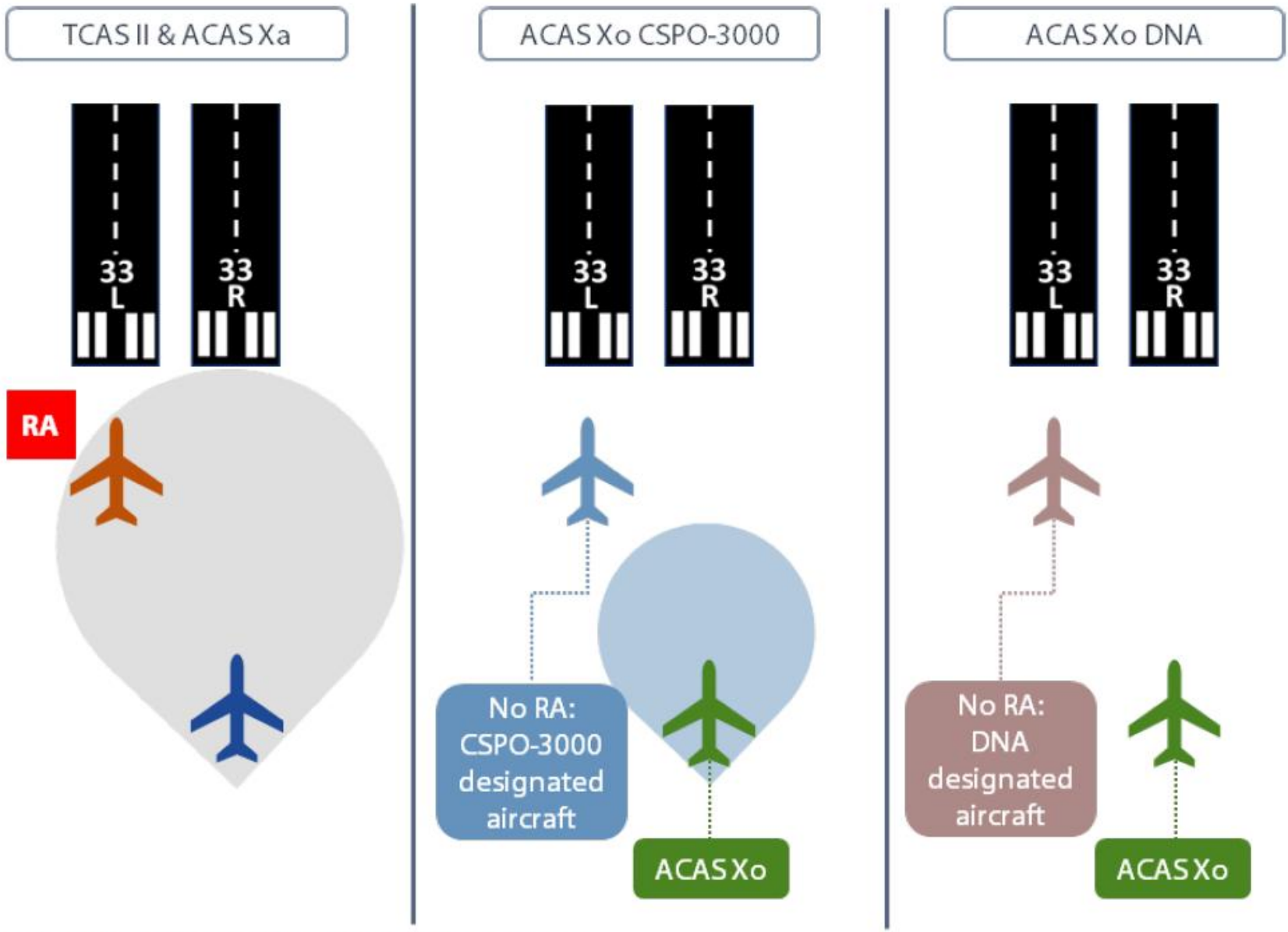


Figure 54: ACAS Xo DNA and CSPO-3000 use cases.

ACAS X_u

EMBRY-RIDDLE
Aeronautical University

ACAS Xu

- ACAS X standard for Uncrewed Aircraft Systems (normal category)
- Issues resolution advisories only.
- Safety net for prevention of Near Mid-Air Collisions and Mid-Air Collisions
- Alerts / Guidance: Rule-based given encounter and tracking of aircraft and local intruders.
  - Unlike ACAS Xa's look up table approach.
- Functions include ensuring
  - Collision Avoidance (CA)
  - Remain Well Clear (RWC)
- Works for cooperative intruders (with transponders or ADS-B) and non-cooperative intruders.
  - Active sensors can be used for non-cooperative systems or under some cooperative encounters.
  - e.g. air-to-air radar meeting RTCA DO-360

ACAS Xu Alert levels from RTCA DO-365

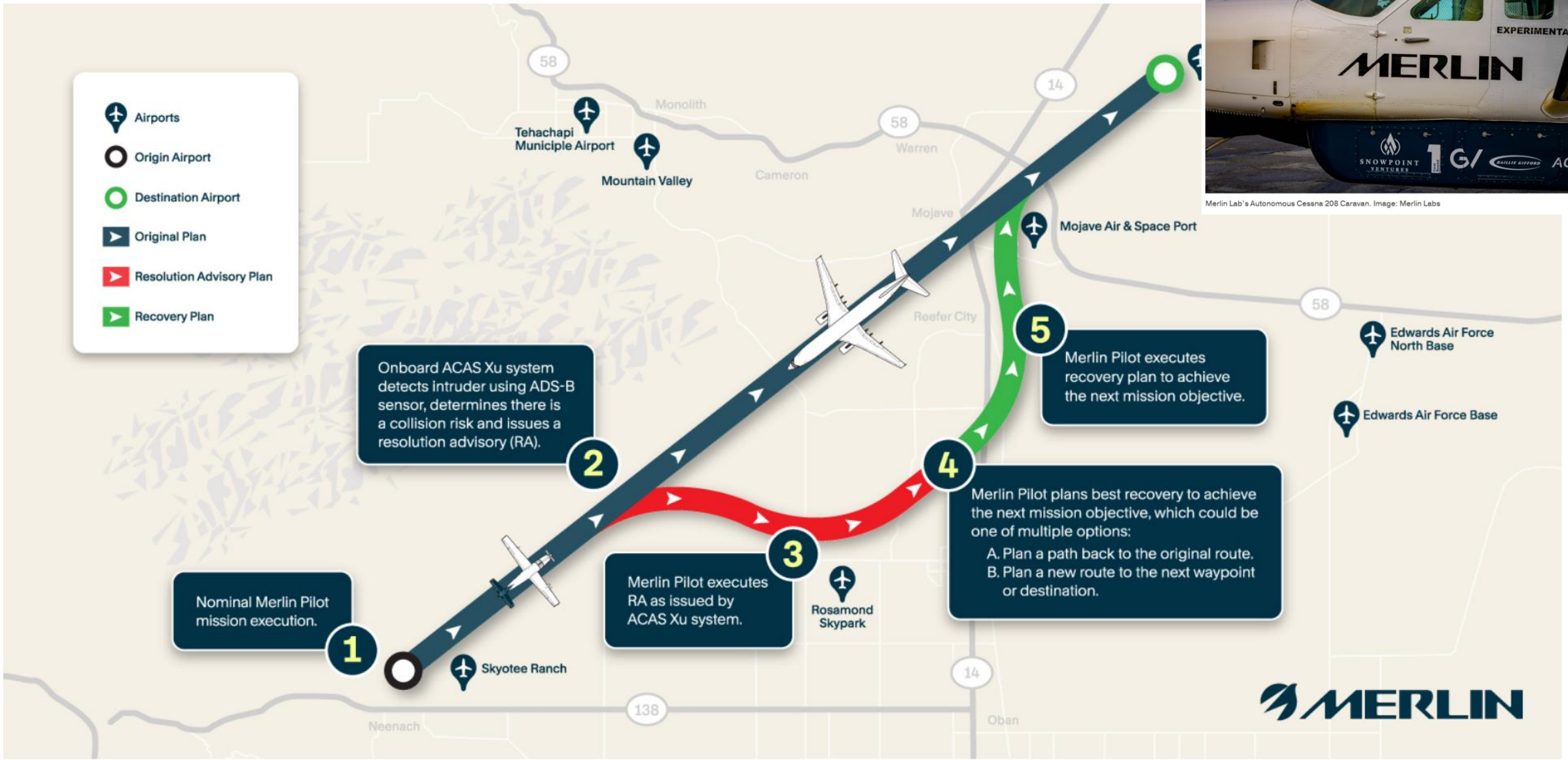
TABLE 2. DAA WELL CLEAR ALERTING PARAMETERS

DWC Alerting Criteria			Units	Preventive Alert	Corrective Alert	Warning Alert
En Route DWC	*Hazard Zone (HAZ)*	τ^*_{mod}	seconds	35	35	35
		HMD*	feet	4,000	4,000	4,000
		h*	feet	700	450	450
	Minimum Average Time of Alert (seconds prior to HAZ)		seconds prior to HAZ	55	55	25
	Late Threshold (seconds prior to HAZ)		seconds prior to HAZ	20	20	15
	Early Threshold (seconds prior to HAZ)		seconds prior to HAZ	75	75	55
Terminal DWC	*Hazard Zone (HAZ)*	τ^*_{mod}	seconds	N/A	N/A	0
		HMD*	feet			1,500
		h*	feet			450
	Minimum Average Time of Alert (seconds prior to HAZ)		seconds prior to HAZ	N/A	N/A	45
	Late Threshold (seconds prior to HAZ)		seconds prior to HAZ			30
	Early Threshold (seconds prior to HAZ)		seconds prior to HAZ			55

EMBRY-RIDDLE Aeronautical University

Sagetech Demo of ACAS-X for UAS/AAM DAA

ACAS Xu Testing : Merlin Labs and MIT Lincoln Laboratory



Merlin Lab's Autonomous Cessna 208 Caravan. Image: Merlin Labs

Merlin Flight Plan Map. Image: Merlin Labs.

[Detect and Avoid Goes Autonomous - Inside Unmanned Systems]

Comparison of alerting (TCAS II and ACAS Xa vs. ACAS Xu)

Differences with ACAS Xa or Xo

- Provides remain well clear function
- Resolution advisories in both the vertical and **horizontal planes**
- Non-cooperative surveillance can support the resolution advisory logic.
- Automation enables ACAS Xu vehicles to automatically respond to a RA and then return to course once resolved (optional)
  - Or Remote Pilot can respond to RA and return to course.
- RAs can be issued based only on ADS-B input
  - Condition: active surveillance must confirm the ADS-B values for position, altitude, etc.
  - Failure to verify ADS-B message results in only using the active sensing method.

Table 18: Key alerting differences between ACAS II and ACAS Xu alerting

	TCAS II & ACAS Xa	ACAS Xu
TAs	Issued	Not issued
RAs against ADS-B only threats	Not issued	Issued
Corrective RAs required vertical rate	1500 ft/min.	1000 ft/min.
Strengthening RA required vertical rate	2500 ft/min.	2000 ft/min.
Preventive RAs annunciation	*"Monitor vertical speed"*	*"Traffic, monitor"*
Weakening preventive RAs annunciation	*"Monitor vertical speed"*	No annunciation
Horizontal RAs	Not issued	Issued, including blended RAs
RA termination	*"Clear of conflict"*	No annunciation

Horizontal Maneuver Guidance using NASA's DAIDALUS

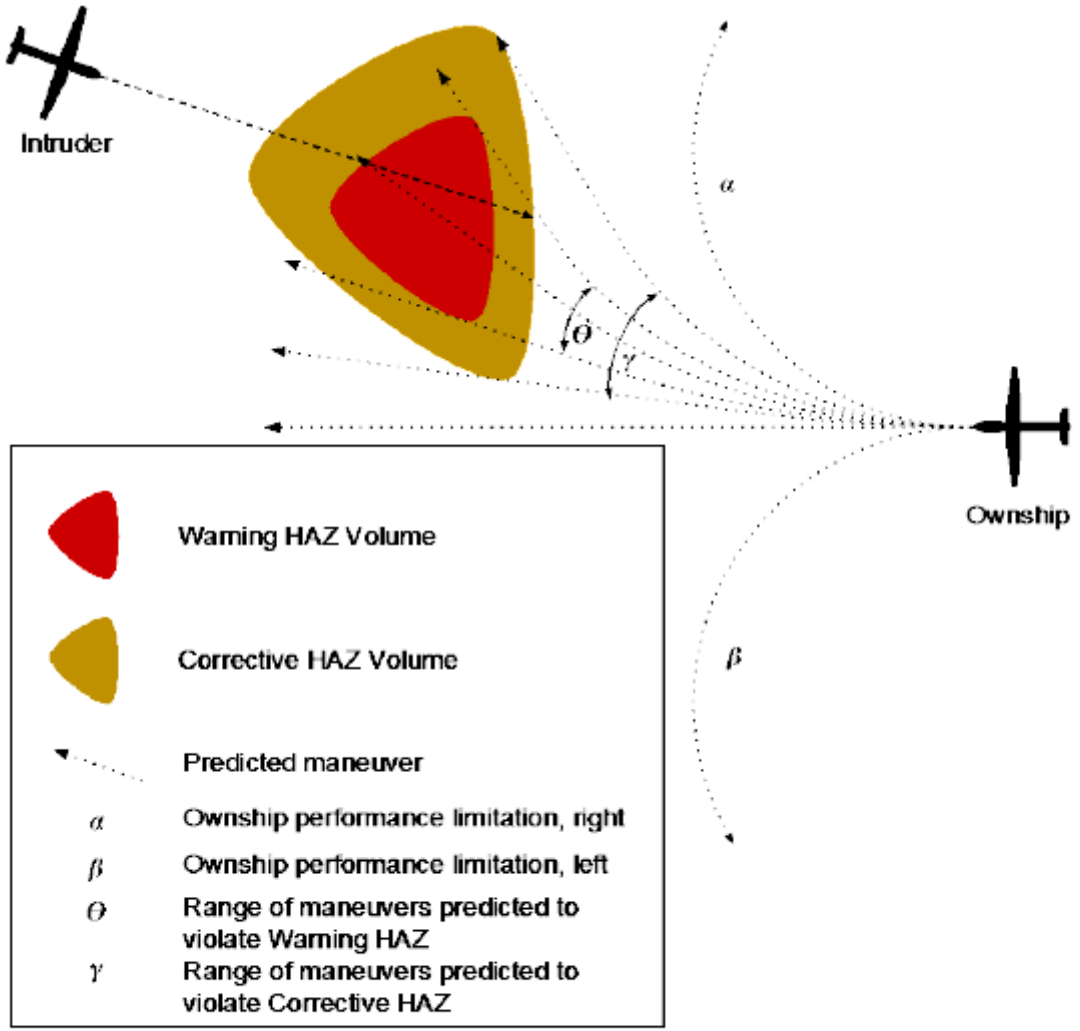


Figure G-3 Concept of DAIDALUS Horizontal Maneuver Guidance to Maintain DWC

Horizontal Resolution Advisories

- Similar approach to vertical Ras
- Display shows the headings that are needed to seek goal as well as headings that can lead toward a collision.
- Blended RA specify BOTH horizontal and vertical changes to mitigate a flight risk.

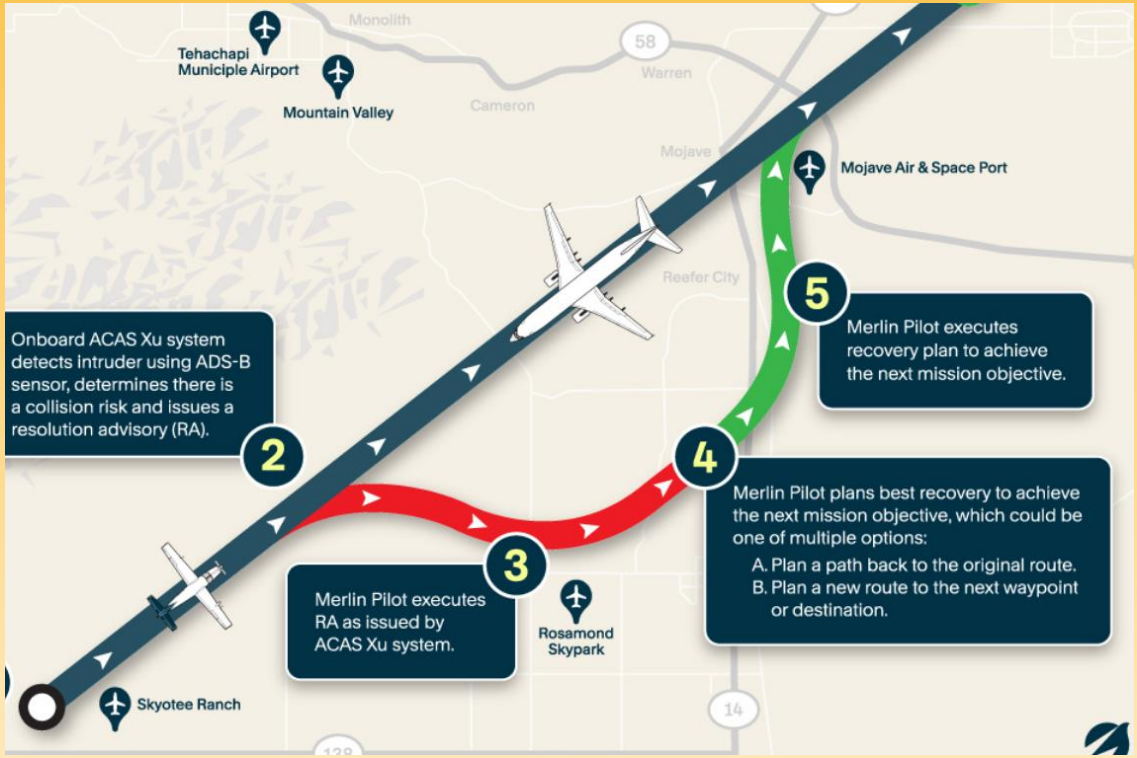
Figure 55: Example of heading display with a horizontal RA.

In the above situation, the pilot on a 0 degree heading must turn left toward 290 degrees (approx.)

Anunciation “turn left”

Handling Alerts

- There are two approaches to addressing alerts
  - Remote Pilot in Command – performs the maneuver and then flies the vehicle back onto course once clear of conflict.
  - Automated flight
    - Onboard software commands the flight control system to execute the appropriate evasive maneuver followed by a Return-to-Course behavior.
    - These automatic responses are not part of ACAS Xu, but can be triggered by ACAS Xu.



Aircraft performance for RA Compliance Requirements

Vertical

- Initial RA:
  - Maneuver within 5 seconds of RA
  - Vertical acceleration of $\frac{1}{4}$ g
- Reversal RA
  - Maneuver within 2.5 seconds of reversal RA
  - Vertical acceleration of $\frac{1}{3}$ g
- Maneuver
  - Vertical Rate for climb and descend are +/- 1000 ft/min
  - Increase Climb or Descend are +/- 2000 ft/min

Horizontal

- Initial RA:
  - Maneuver with 5 seconds of RA
- Reversal RA
  - Maneuver within 2.5 seconds of reversal RA
- Turn rate:
  - 3 deg/s turn rate
  - 1 deg/s^2 acceleration
- If the performance cannot be achieved, ACAS Xu can attempt to compensate for the underperformance to ensure safety.

Coordination w/ ACAS Equipped Intruder(s)

- Both aircraft w/ RA Capability
  - Each system produces an RA
  - Through an interrogation message, each vehicle can assess and reply to the other whether the proposed RA will be compatible.
  - i.e. The following coordination approach is followed (from the perspective of Ownship)
    - Ownship will share its plan.
    - Intruder will assess the RA sent from ownship to determine if they are compatible (will not collide or NMAC), and respond with approval/rejection. And vice-versa.
    - Once both agree with the other's plan, they take action.

Questions